WHO IS AT RISK OF MENTAL ILLNESS?

Rebecca Cheng, Judy Gong, Yawen Hou
Applied Data Analysis @ EPFL, Switzerland

canADA

The Dataset

In this project, we will mainly look at the prevalence of mental disorders over the world. We will also look at its impact (quantified with Disability-Adjusted Life Year (DALY)) and its correlation with Life Satisfaction, Education and Employment with chosen environment. Our data comes from the Global Burden of Disease (GBD) database. This is a database maintained by The Institute for Health Metrics and Evaluation (IHME), part of the University of Washington, and provide rigorous data about the world population health. Their dataset provide the various causes of illness, including physical and mental illness, from 1990-2017. All the data of this project have been retrieved from the GBD, which we cleaned and analyzed using techniques learnt from the EPFL ADA course.

The dataset can be found here: http://ghdx.healthdata.org/gbd-2017

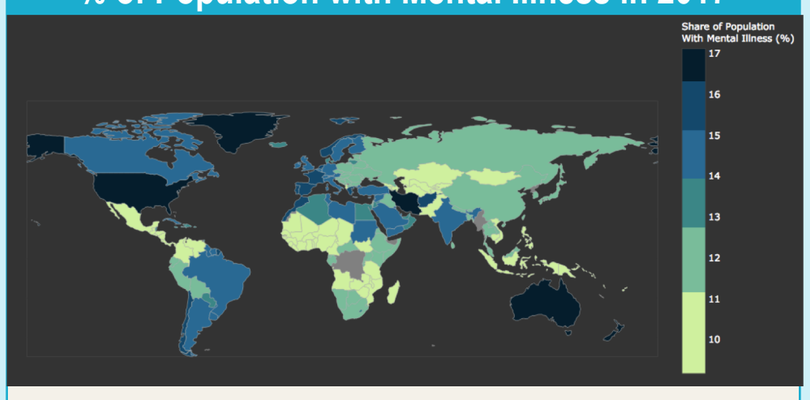
Introduction

6 Facts from the World Health Organization (WHO)

Fact 1: About half of the mental disorders begin before the age of 14.
Fact 2: Neuropsychiatric disorders (eg. Eating Disorders, Depression, Bipolar disorder, etc.) are among the leading causes of worldwide disability in young people.
Fact 3: Around 20% of world's children and adolescents have mental disorders or problems.
Fact 4: About 23% of all years are lost because of disability by mental and substance use disorders, make it the leading cause of disability worldwide.
Fact 5: War and disasters have a large impact on mental health and psychosocial well-being.

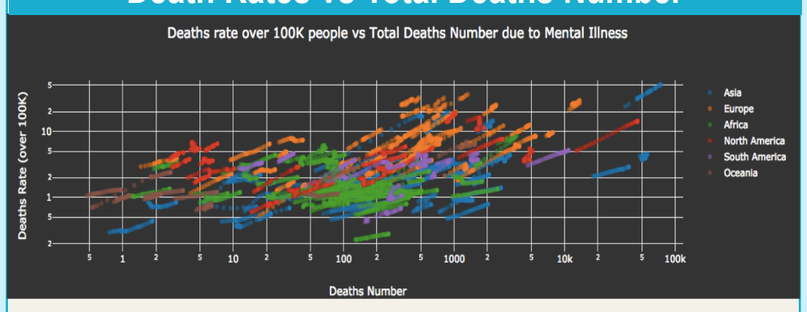
Our goal for this project is to develop insights on key risk factors of mental illness

% of Population with Mental Illness in 2017



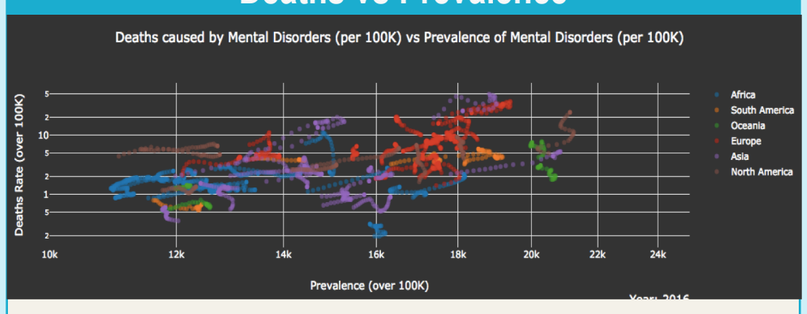
As can be seen in the map above, the countries' share of population with mental illness consistently falls between 9% to 17%. The countries that the highest mental illness prevalence include: United States, Australia, Greenland and Iran. United State's dark period falls between 1997 – 2009, which could be correlated to the 9/11 attack and financial crisis that occurred during this period.

Death Rates vs Total Deaths Number



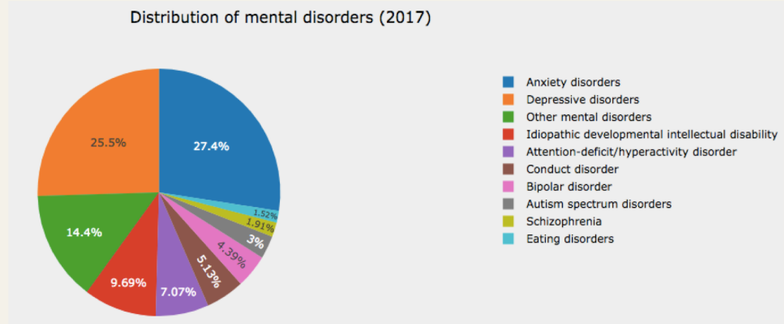
Mental disorders related deaths occur throughout the world, affecting individuals of all nations, cultures, religions and genders. We can see from the graph below that the prevalence of mental disorders does not show a direct relationship with the death rates. We also realized that the increase in death number is not only caused by the increase of the population, since the death rate (number of death over 100 000 people) also increases over years. In fact, statistics show that countries with highest mental disorder related death rates in the world are incredibly diverse.

Deaths vs Prevalence



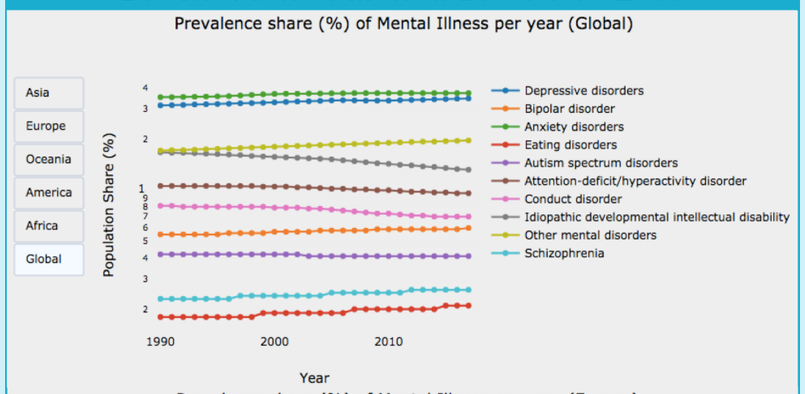
Substance use disorders (drug and alcohol use) contribute to more than 100 k deaths per year. Specifically, Russia, Australia, the United States and Nordic countries appear to have the highest direct deaths rates reported from substance use disorders in 2016. China, India and the United States had the highest direct deaths from anorexia and bulimia nervosa, with more than 60 deaths each in 2016.

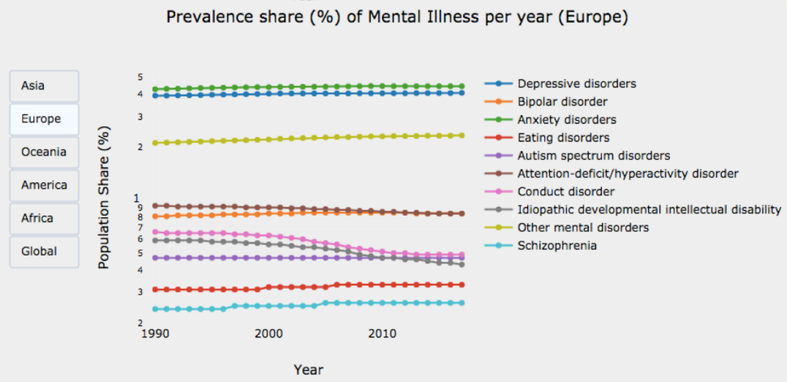
Distribution of Mental Disorders 2017

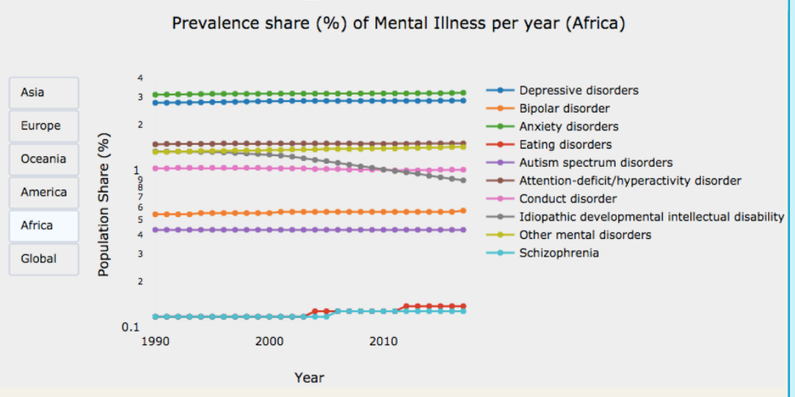


Among all the people having mental disorders, more than 50% are victims of either anxiety or depressive disorders. On a global scale, only very few people have schizophrenia and eating disorders, despite the increasing cases of obesity and the general fashion trend that values skinniness.

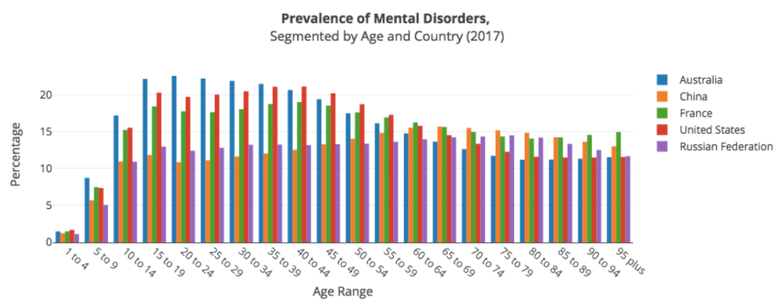
Distribution of Mental Disorders 2017

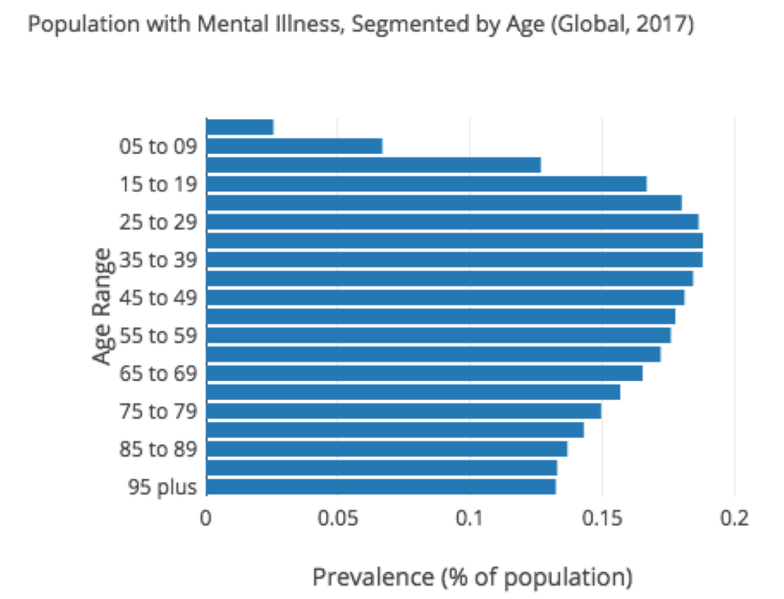






Distribution of Mental Disorders by Age

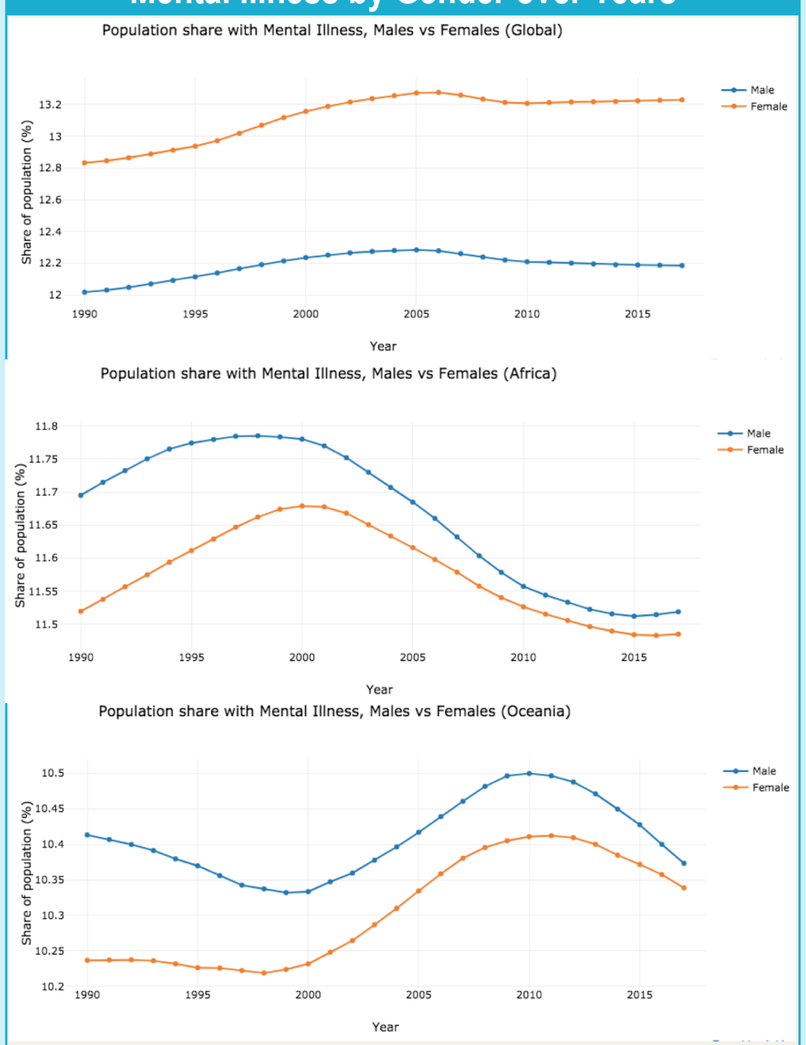




Every age group is affected by depression. Certain times of life can make us more susceptible. During infancy, 2.5% of this population starts to experience mental disorders, mainly anxiety disorders. The prevalence of mental disorders increases significantly for youth (12.6%) and young adolescents (16.6%). Approximately 1 in 5 adults globally, or 18.5% experiences mental illness in a given year.

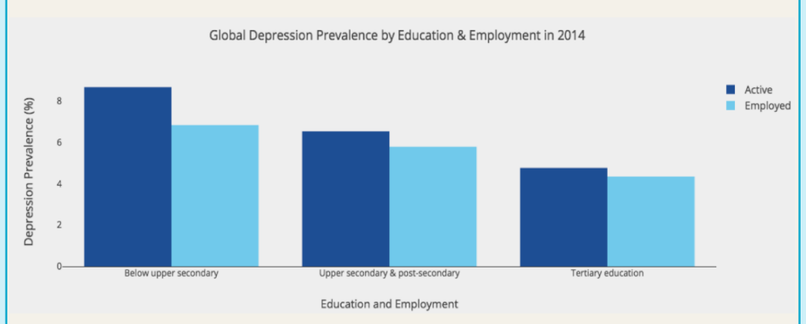
Mental Disorder by Gender + Country in 2017

Mental Illness by Gender over Years



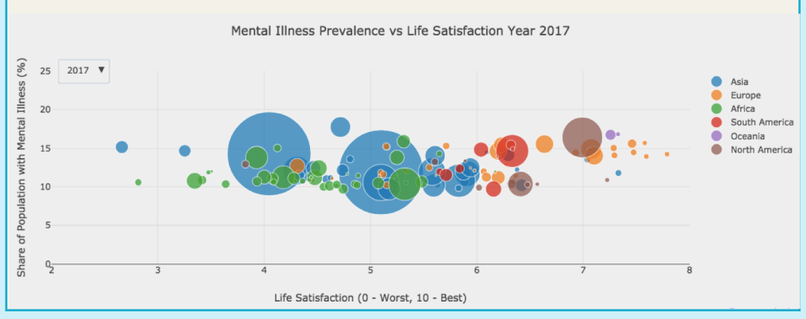
We notice that on a global trend, female tend to be more affected by mental disorders. This fact has been confirmed by a study done by Gulland (2016). This is true for America, Europe and Asia, where the share of female population having mental disorders are generally 0.4-2% higher than the share of male population. We can also observe that the rate of people having mental disorders increases over time. We observe the opposite in Africa and Oceania.

Depression by Education/Employment 2014



We decided to perform a hypothesis test to test the difference in depression prevalence between the active and employed groups. Using a two-sided t-test, we arrived at a final p-value of 0.067. Using a 10% significance level, we rejected our null hypothesis and concluded that there is a significant difference so unemployed people tend to be more depressed.

Mental Illness vs Life Satisfaction



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